🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.5

Explore Numerical Expressions with Exponents

Lesson 6-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

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SOUND STUDIO MISSION

The Doubling Mixboard

You are the lead sound engineer at Amplitude Studios. Every time you flip a switch on the new mixboard, the volume doubles — that is the power of exponents. The headline act goes on in one hour, and you must master powers before the speakers can be safely dialed in.

My Notes
Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write and evaluate numbers in exponent form using a base and a power.

Language Objective: I can explain my work using the words exponent, base, power, and evaluate.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Exponent | Base | Power | Evaluate

👆 *Tap any word to see its meaning and a picture.*

1. The small raised number that shows how many times the base is multiplied is the ________.

2. The number being multiplied repeatedly in a power, like the 3 in 3^4, is the ________.

3. A number written with a base and an exponent, like 3^4, is a ________.

4. To find the value of an expression, like $2^3 = 8$, is to ________ it.

Write About the Math

How large is the file after 5 layers? Why does the file size grow so quickly?
¿De qué tamaño es el archivo después de 5 capas? ¿Por qué crece tan rápido el tamaño del archivo?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Exponent** *Exponente*
- [ ] **Base** *Base*
- [ ] **Power** *Potencia*
- [ ] **Evaluate** *Evaluar*

Model: *The exponent counts how many times we multiply, not what we add.* — Students connect each switch flip to one more factor of 2 and explain that $2^3 = 2 \times 2 \times 2 = 8$, distinguishing repeated multiplication from repeated addition (which would give 6).

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **After 5 layers the file is ____ MB because $4 \times 2^5 = 4 \times$ ____ = ____. The file grows quickly because each layer ____ the size.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.
Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Exponent
2. **Notes 2:** Base
3. **Notes 3:** Power
4. **Notes 4:** Evaluate
5. **Watch for this mistake:** *A common mistake in Powers and Exponents is multiplying the base by the exponent instead of using the base as a repeated factor — for example, thinking* $2^3 = 2 \times 3 = 6$ *instead of* $2^3 = 2 \times 2 \times 2 = 8$. *Before you submit, rewrite the power as repeated multiplication first, then multiply step by step.*
6. **Try It 1:** B. 7^3 — *The base 7 is used as a factor 3 times, so the power is* 7^3. $7^3 = 7 \times 7 \times 7 = 343$.
7. **Try It 2:** D. 64 — $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$. *The exponent 6 counts how many times 2 is a factor.*